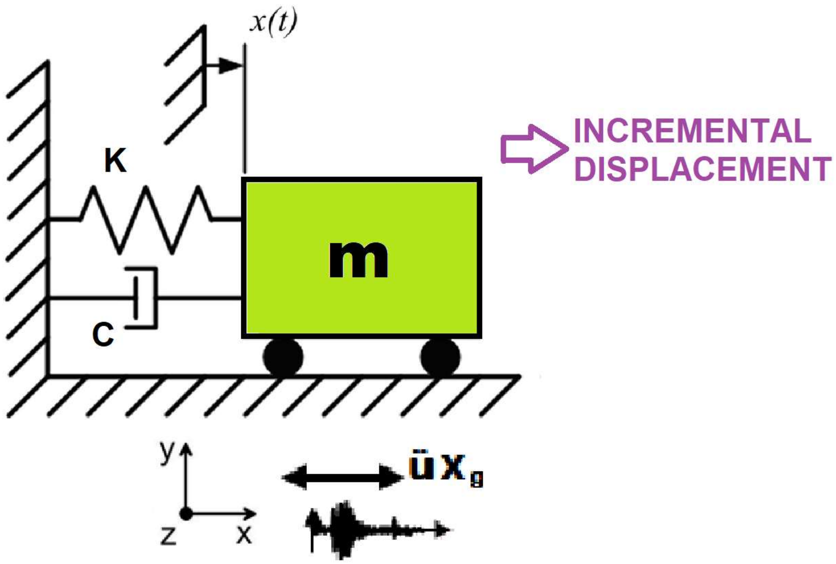
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF OPTIMUM STRUCTURAL DUCTILITY RATIO FROM ENERGY DISSIPATION CAPACITY INDEX USING FINITE-DIFFERENCE NEWTON ITERATION

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

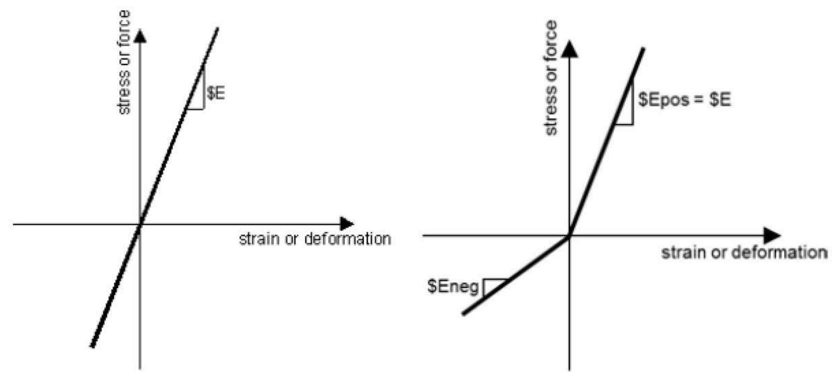
$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

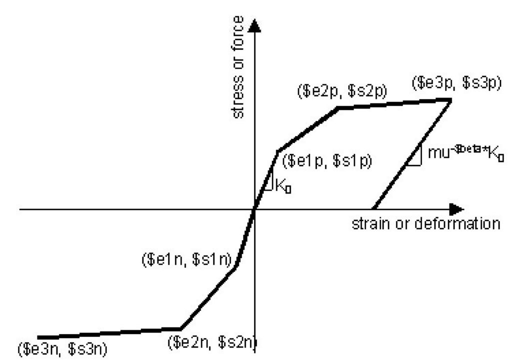
$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

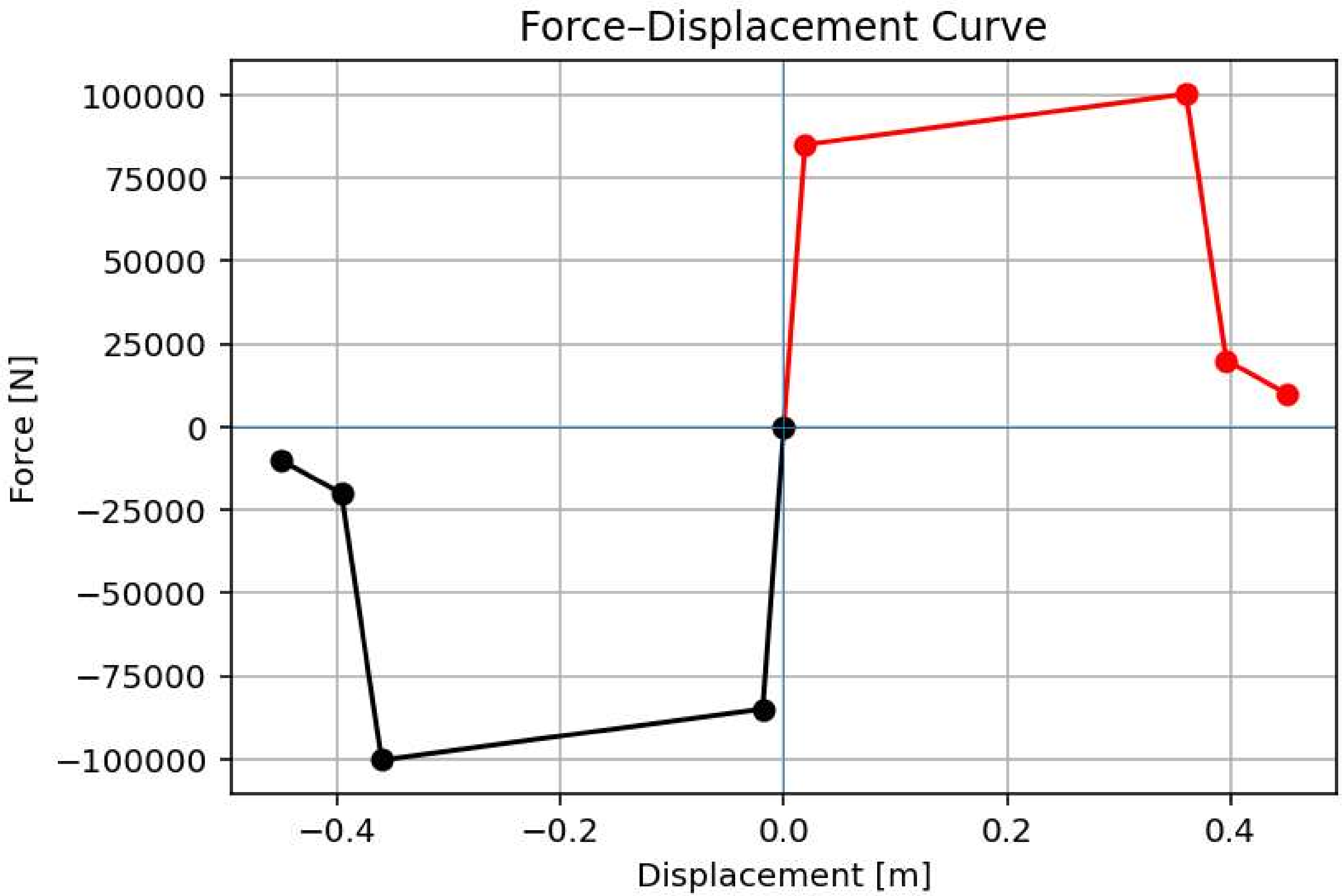
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



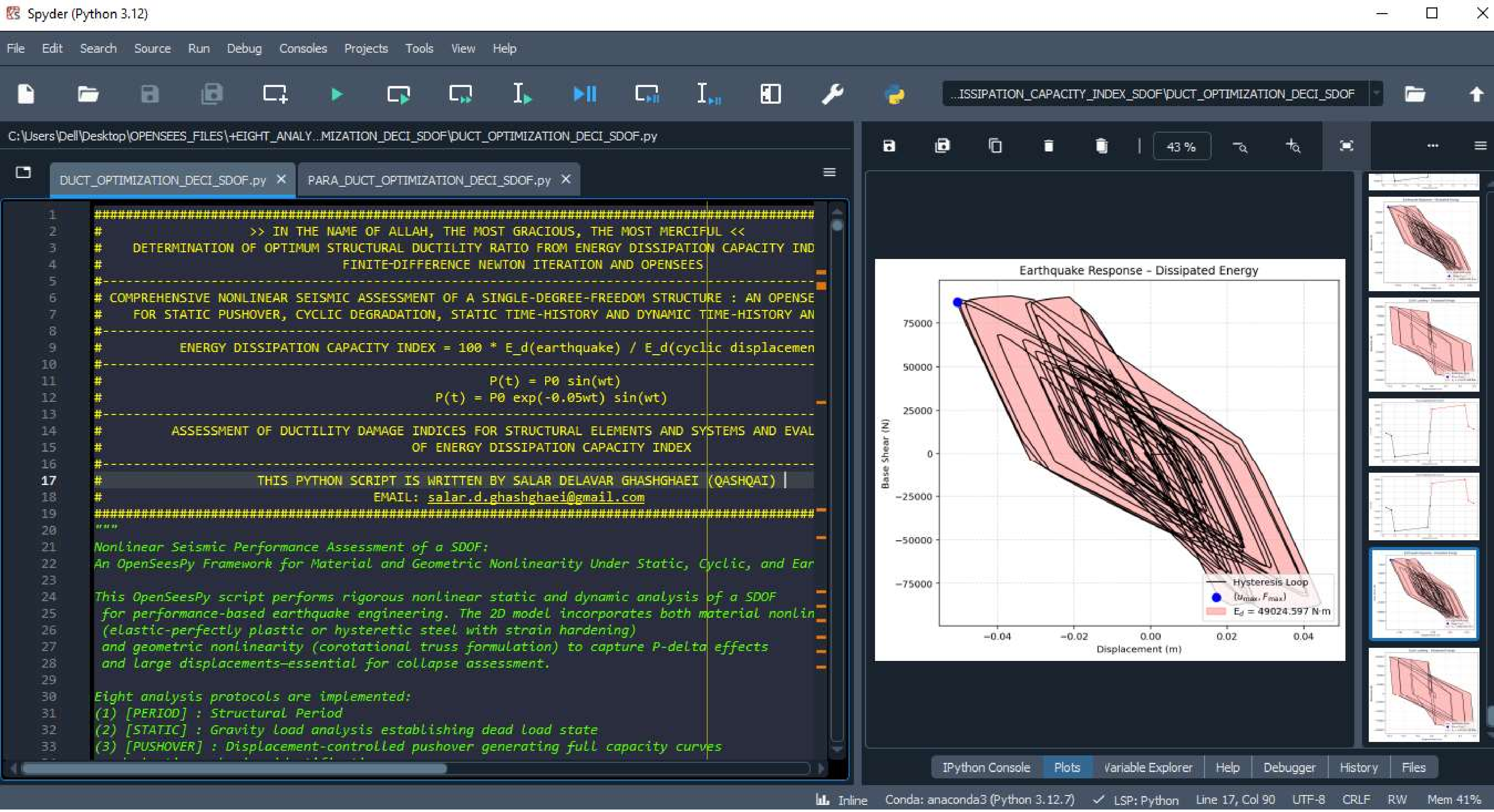
ELASTIC MATERIAL



INELASTIC MATERIAL



Force–Displacement Curve
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement [m]


Spyder (Python 3.12)
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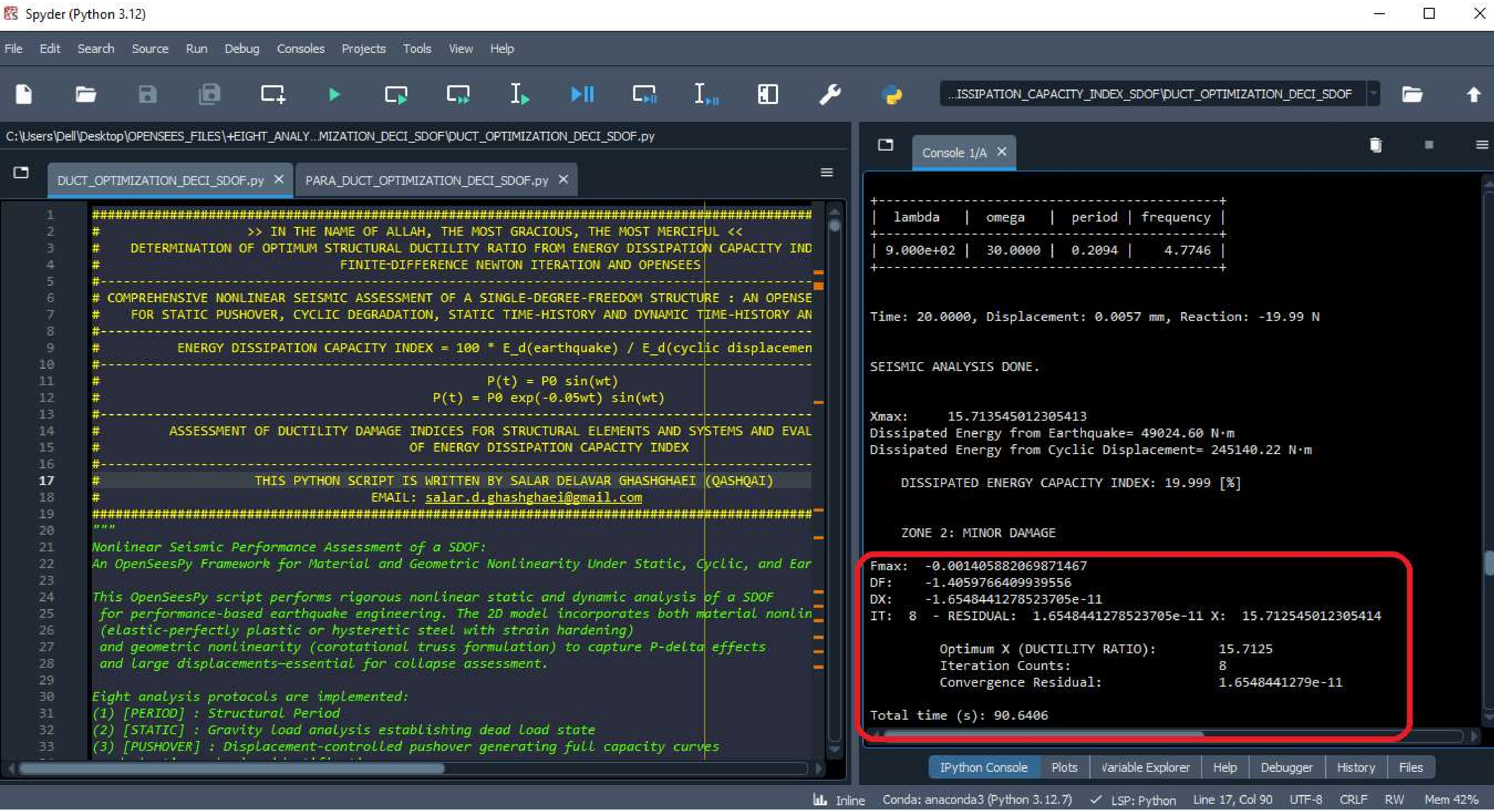
FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM STRUCTURE : AN OPENSE
FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY AN
ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacemen
P(t) = P0 sin(wt)
P(t) = P0 exp(-0.05wt) sin(wt)
ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVAL
OF ENERGY DISSIPATION CAPACITY INDEX
THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
"""
Nonlinear Seismic Performance Assessment of a SDOF:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Ear
This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a SDOF
 for performance-based earthquake engineering. The 2D model incorporates both material nonlin
 (elastic-perfectly plastic or hysteretic steel with strain hardening)
 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
 and large displacements—essential for collapse assessment.
Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
43 %
Earthquake Response - Dissipated Energy
Base Shear (N)
Displacement (m)
Hysteresis Loop
(u_{max}, F_{max})
E_d = 49024.597 N·m
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 17, Col 90 UTF-8 CRLF RW Mem 41%

OPTIMIZATION PROBLEM FOR ENERGY DISSIPATION CAPACITY INDEX (EDCI)


Spyder (Python 3.12)
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DUCT_OPTIMIZATION_DECI_SDOF.py
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Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
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(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
Console 1/A
| lambda | omega | period | frequency |
| 9.000e+02 | 30.0000 | 0.2094 | 4.7746 |
Time: 20.0000, Displacement: 0.0057 mm, Reaction: -19.99 N
SEISMIC ANALYSIS DONE.
Xmax: 15.713545012305413
Dissipated Energy from Earthquake= 49024.60 N·m
Dissipated Energy from Cyclic Displacement= 245140.22 N·m
DISSIPATED ENERGY CAPACITY INDEX: 19.999 [%]
ZONE 2: MINOR DAMAGE
Fmax: -0.001405882069871467
DF: -1.4059766409939556
DX: -1.6548441278523705e-11
IT: 8 - RESIDUAL: 1.6548441278523705e-11 X: 15.712545012305414
Optimum X (DUCTILITY RATIO): 15.7125
Iteration Counts: 8
Convergence Residual: 1.6548441279e-11
Total time (s): 90.6406
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 17, Col 90 UTF-8 CRLF RW Mem 42%

Dissipated Energy from Earthquake= 49024.60 N·m
Dissipated Energy from Cyclic Displacement= 245140.22 N·m

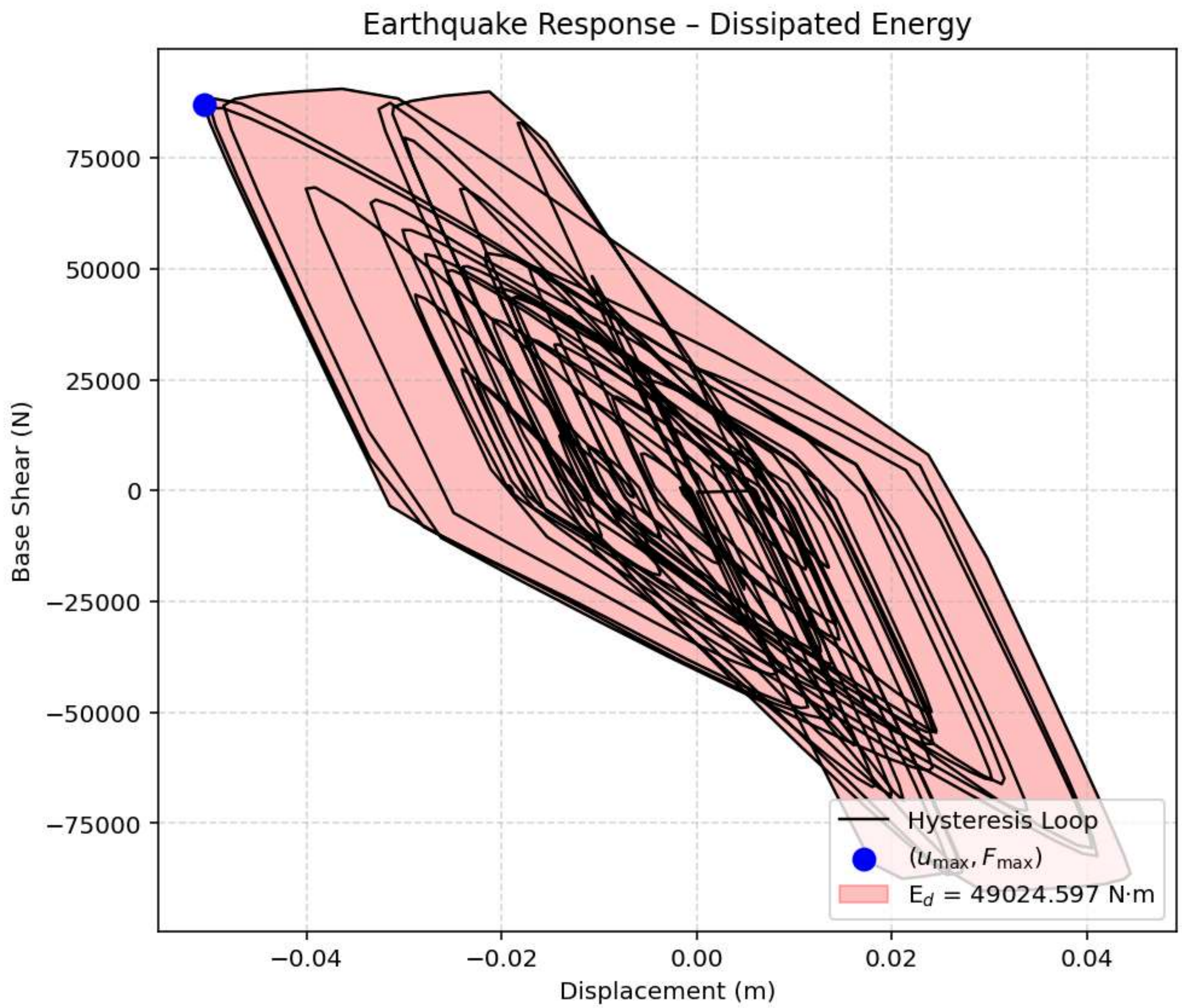
DISSIPATED ENERGY CAPACITY INDEX: 19.999 [%]

ZONE 2: MINOR DAMAGE

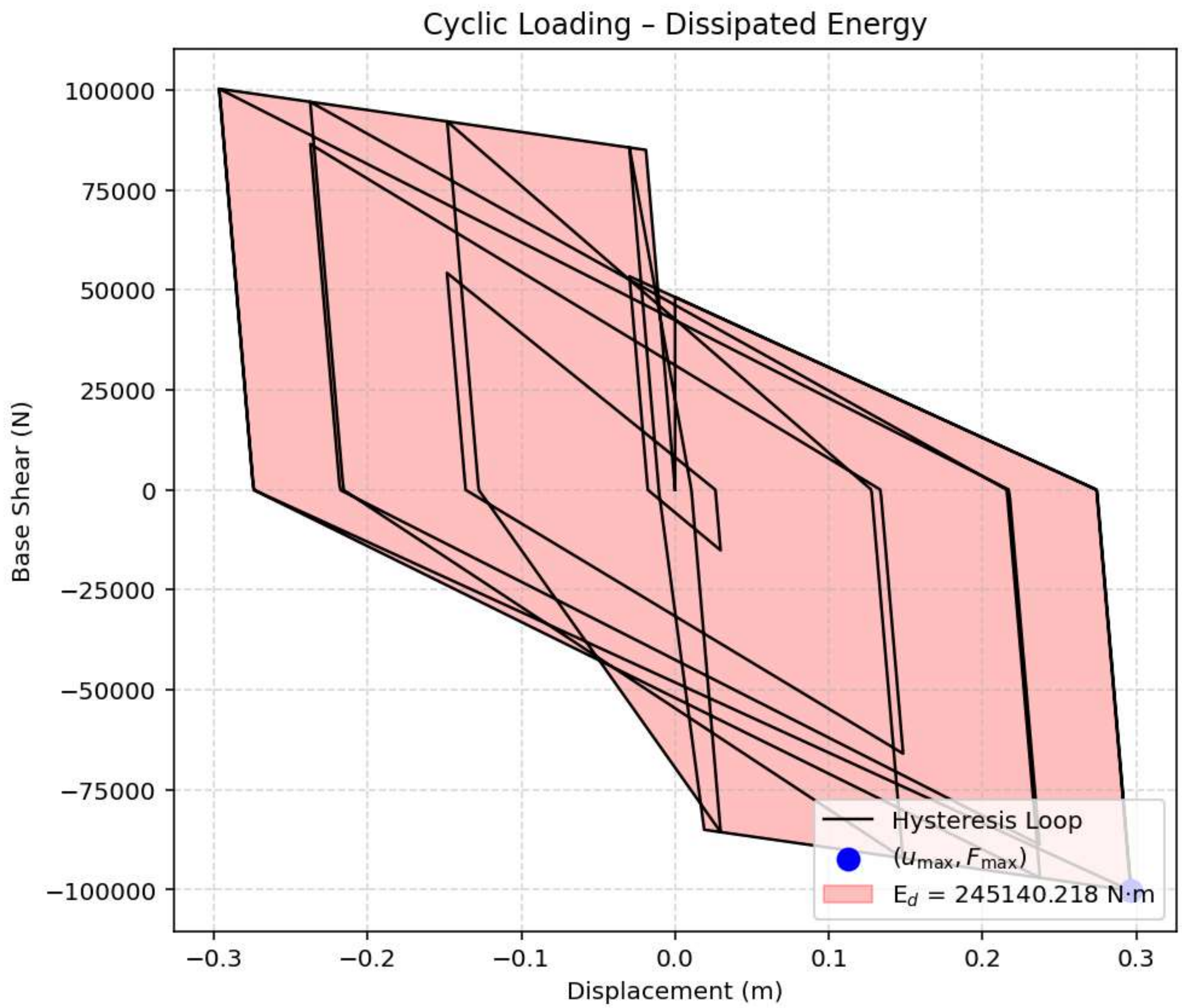
Fmax: -0.001405882069871467
DF: -1.4059766409939556
DX: -1.6548441278523705e-11
IT: 8 - RESIDUAL: 1.6548441278523705e-11 X: 15.712545012305414

Optimum X (DUCTILITY RATIO): 15.7125
Iteration Counts: 8
Convergence Residual: 1.6548441279e-11

Total time (s): 90.6406



Earthquake Response – Dissipated Energy
Base Shear (N)
75000
50000
25000
0
−25000
−50000
−75000
−0.04
−0.02
0.00
0.02
0.04
Displacement (m)
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 49024.597 N·m



Cyclic Loading – Dissipated Energy
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Base Shear (N)
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement (m)
Hysteresis Loop
($u_{\max}$, $F_{\max}$)
E$_d$ = 245140.218 N·m